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Comparing Problem-Specific and Mean Hidden-State Interventions for Context-Reconstructed Reasoning

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Slides (PDF)

LLMs can fail after a context change

Original (ORG)

A man fell from a 100-rung ladder, but was unhurt. How?





Correct answer: the lowest rung

Context-Reconstructed (CR)

A man fell from a 50-story building, but was unhurt. How?





Correct answer: the lowest floor

Different contexts, the same underlying solution structure.

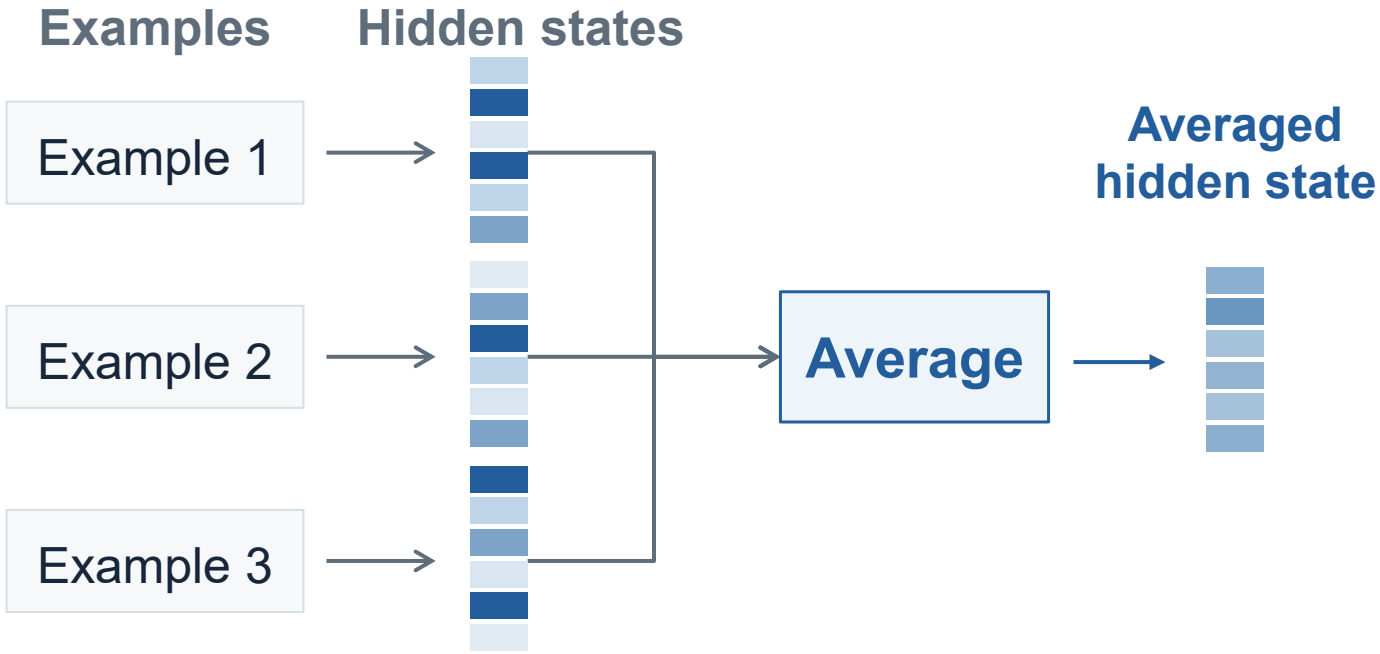
Illustrative example. Schematic responses, not measured.

Related Work: Interventions with averaged states

Hidden states

Numerical vectors computed at each token and layer

- Changing hidden states can alter model behavior. [6–8]
- Many interventions use average directions. [6–8]



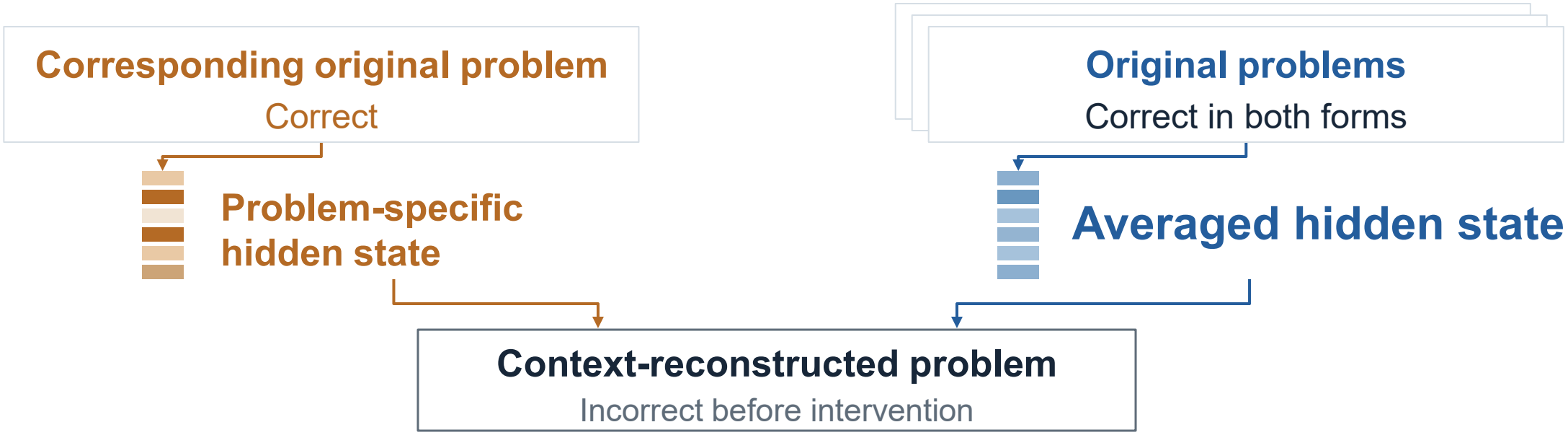
Potential limitation

Averaging may dilute problem-specific information needed to recover a reasoning failure.

Li et al. (2023), Liu et al. (2024), Todd et al. (2024)

Research Question

Does a problem-specific hidden state recover reasoning failures more effectively than an averaged hidden state?



Main finding

Problem-specific states give higher peaks in all four models

Problem pairs and answer evaluation

BrainTeaser Sentence Puzzle (BTSP) — **209 problem pairs**

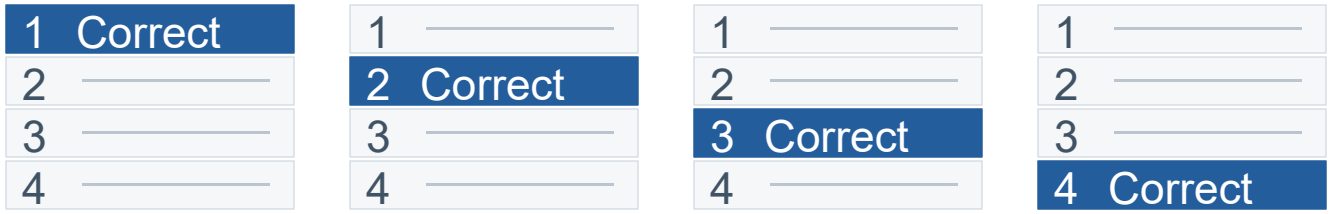
Four-choice lateral-thinking word puzzles

Original (ORG) → **Context-Reconstructed (CR)**

Input	Each model	Output
Question + four answer options →	• Gemma 2 9B IT • Gemma 3 12B IT • Gemma 4 E2B IT • Gemma 4 E4B IT	→ Answer number 1–4

Zero-shot. Select the highest candidate log-likelihood.

Same correct option, four positions (ORG / CR separately)



Answer 1 ✓ Answer 2 ✓ Answer 3 ✓ Answer 4 ✓

→ All 4 answers correct

Problem correct

Any error: incorrect

Failures after a context change

Letter order: ORG / CR C = Correct I = Incorrect

Targets for recovery

CI group = Correct / Incorrect

Sources for averaging

CC group = Correct / Correct

Problem pairs, classified separately for each model

Model	CI group	CC group
Gemma 2 9B IT	**40**	113
Gemma 3 12B IT	**38**	94
Gemma 4 E2B IT	**35**	11
Gemma 4 E4B IT	**32**	99

Choosing hidden states

Problem-specific (Pair-CI)

Hidden state from the corresponding ORG problem

Averaged (Mean-CC)

Average of ORG hidden states from the CC group

Additional comparisons	ORG source	Comparison purpose
Mean-CI	Averaged ORG hidden state from other CI problems	Source-group membership
Surface-similar-CI / Surface-similar-CC	ORG with highest TF-IDF similarity to CR	Lexical similarity
Random-CI / Random-CC	One randomly selected ORG	Arbitrary source choice

CI-source controls use Gemma 4 models and exclude the corresponding ORG problem.

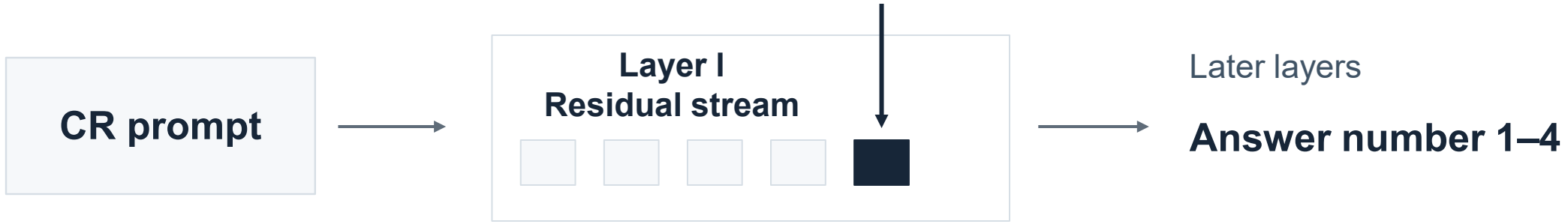
Hidden-state intervention

Source hidden state

Average hidden states over four option-slot variants per ORG

Norm-matched source

Match the norm of the target problem's hidden state



$$\tilde{h}_{CR} = h_{CR} + \alpha(\hat{h}_{src} - h_{CR})$$

h_{CR}: target state $\hat{h}_{src}$: norm-matched source

- α = 1.0

  Replace with the norm-matched source

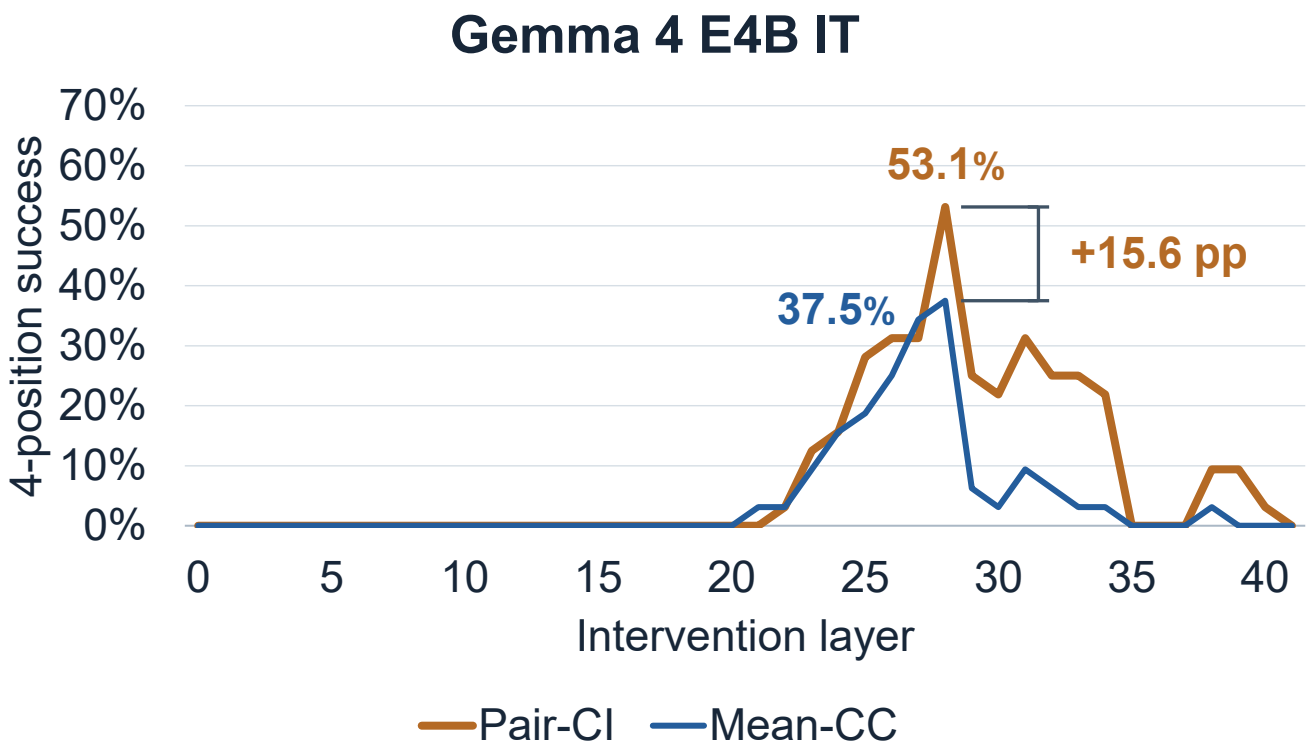
- Final prompt position
- All layers, one at a time

4-position success rate

CI group CR targets correct in all four variants (%)

Higher peak recovery in all four models

α = 1.0 Gaps in percentage points (pp)



Peak recovery

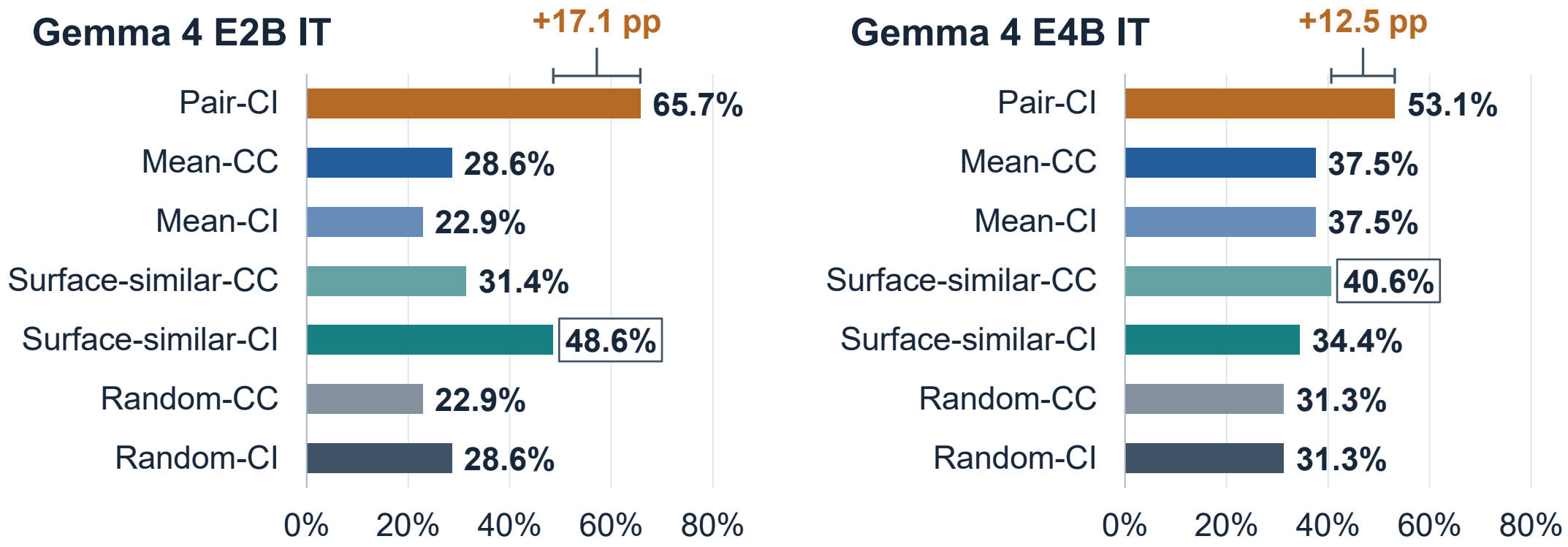
Model	Pair-CI	Mean-CC
Gemma 2 9B IT	**22.5%**	15.0%
Gemma 3 12B IT	**15.8%**	7.9%
Gemma 4 E2B IT	**65.7%**	28.6%
Gemma 4 E4B IT	**53.1%**	37.5%

Peak over layers

Averaging across problems may weaken useful information.

Other sources give lower peaks

Gaps: Pair-CI vs best alternative; peaks over layers at α = 1.0.

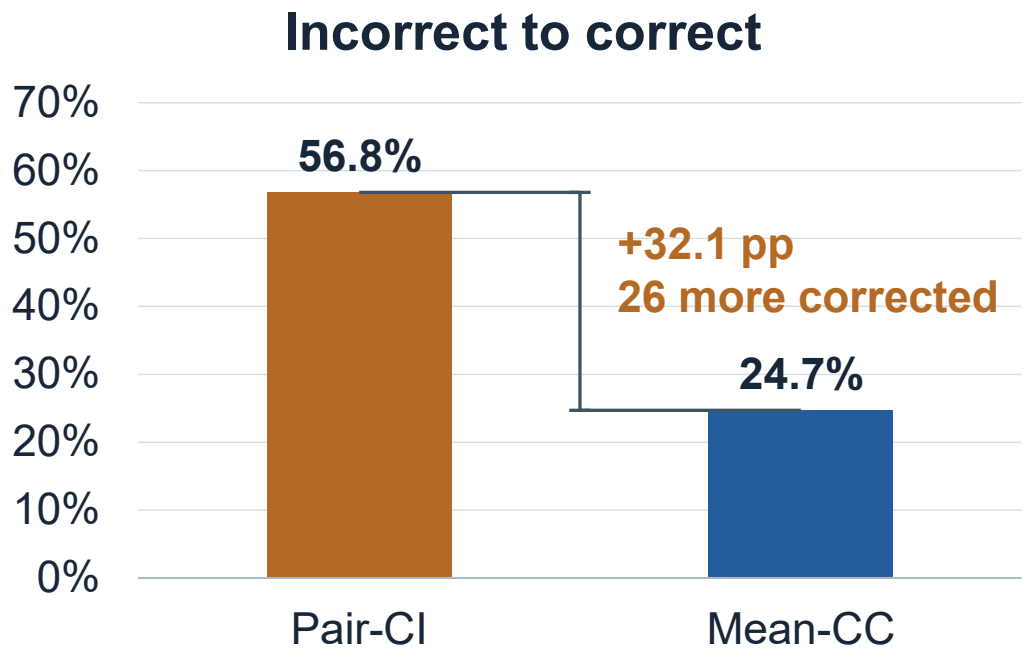


The tested group and lexical controls do not reproduce the Pair-CI peak.

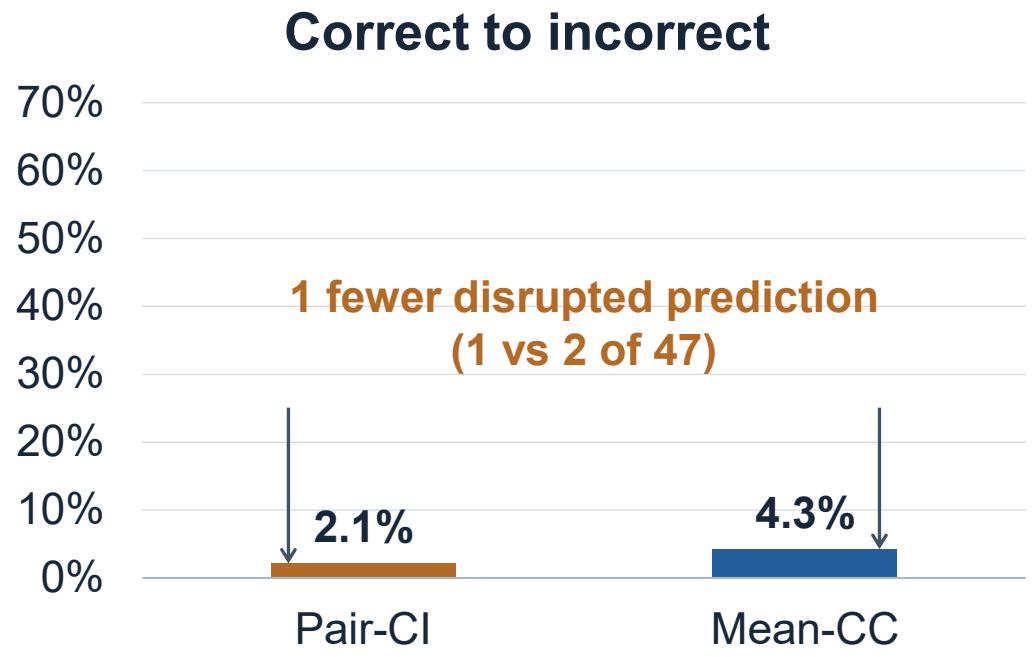
Correcting errors with little disruption

Gemma 4 E4B IT, layer 28, α = 1.0

81 initially incorrect predictions



47 initially correct predictions



At this layer, Pair-CI corrects more errors with little disruption.

Limitations

Meaning	Semantic content and full solution structure remain unidentified.
Scope	One dataset, four models, one failure group. One prompt position and one layer at a time.
Peak estimates	Descriptive layer maxima. No held-out selection or uncertainty estimates.

Answer to the research question

Does a **problem-specific hidden state** recover reasoning failures more effectively than an **averaged hidden state**?

Problem-specific states give higher peak recovery.

All four tested models, α = 1.0 (Pair-CI vs Mean-CC)

Pair-CI on E4B, layer 28: more error correction with little disruption of correct predictions.

Next Identify the useful information and test broader reasoning tasks.